A Novel Multitemporal Image-Fusion Algorithm: Method and Application to GOCI and Himawari Images for Inland Water Remote Sensing

Yulong Guo, Changchun Huang, Yali Zhang, Yuan Li, and Weiqiang Chen

***Abstract*— A spectral–temporal-unmixing-based multitemporal image fusion (MTIF) algorithm is proposed to fuse Geostationary Ocean Color Imager (GOCI) and Himawari images. The algorithm was applied to two data sets. The fusions are quantitatively and qualitatively compared with four widely used algorithms. The results show that the MTIF algorithm performs better using both evaluation indexes and visual comparisons. For optical complex water monitoring, the MTIF-derived chlorophyll-a concentration (C_{chla}) map has better spatial detail and temporal trends compared with the other algorithms. For cloudy images, the MTIF algorithm can estimate part of the undercloud water reflectance information: when there are more than 32.6 cloud-free pixels in the current study area, the MTIF algorithm can successfully recover the under cloud information. The MTIF algorithm has great potential to advance the monitoring of optical complex inland water.**

***Index Terms*— Geostationary Ocean Color Imager (GOCI), Himawari, image fusion, optical complex water monitoring.**

I. Introduction

In THE past few decades, inland water pollution has become a major problem due to anthropogenic activities, such as rapid urbanization. Remote sensing provides an effective way to monitor changing inland water bodies. However, unlike open water areas, inland water has more complex temporal, spatial, and optical properties. Therefore, data-source selection in inland water remote-monitoring missions is complicated. Landsat image has sufficient spatial resolution to evaluate the spatial patterns of water quality [1]–[5]. Median spatial resolution images, such as Moderate Resolution Imaging Spectroradiometer (MODIS), Medium Resolution Imaging Spectrometer (MERIS), Sentinel 3A Ocean and Land Colour Instrument (OLCI), etc., are capable of accurately capturing water pigment due to their spectral settings [6]–[11]. Recently launched geostationary satellites, such as Geostationary Ocean Color Imager (GOCI), Spinning Enhanced Visible and Infrared Imager (SEVIRI), and Himawari, offer the opportunity to monitor detailed temporal variation in water quality [12]–[16]. An ideal sensor for inland water monitoring should have high spatial, spectral, and temporal resolutions. However, those advantages cannot be obtained from the same sensor because of hardware limitations: Landsat's long revisit cycle (14 days) cannot keep track of the rapidly changing bio-optical processes of inland lakes. The coarse spatial resolution of MODIS-like images cannot accurately determine water boundaries. The scale effect caused by the low spatial resolution will also influence the water color parameters' inversion. Therefore, image fusion is a useful tool for water quality monitoring [16]–[20].

The current image fusion algorithms can be separated into three types.

1) Spatial and spectral fusion algorithms, such as spectral-transform-based methods such as Hue-Intensity-Saturation (HIS) transform [21], [22], color normalized (CN) [23], principle component (PC) [24], [25], Gram–Schmidt (GS) transform [26], [27], wavelet transform-based algorithms [28]–[32], discrete wavelet transform-based algorithm [33], "à trous" wavelet transform-based algorithm [34], contourlets-based algorithm [35], support value transform-based algorithm [36], and injection model-based algorithms [37]. Wang *et al.* [29] proposed a comprehensive framework—the general image fusion (GIF) method—to categorize, compare, and evaluate these fusion methods. The GIF algorithm indicated that a lot of pan-sharpening algorithms could be seen as the sum up of multispectral images' spectral information and PAN images' spatial-detail information. Additionally, some researchers use the sparse representation method [38]–[40] to solve the pan-sharpening problem. The pan-sharpening algorithms are aiming to merge high spatial resolution images with low spatial resolution multispectral images, or medium-high spatial resolution multispectral images with low resolution hyperspectral images. The fusions are multispectral/hyperspectral images with better spatial details. In the fusion algorithms, only the

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synchronous or quasisynchronous multisource images are fused. That is, the spatial–spectral fusion algorithms ignored the temporal changes in the multisource images.

2) Spatial and temporal fusion algorithms. The Spatial and Temporal Adaptive Reflectance Fusion Model (STARFM) was proposed to fuse Landsat TM and MODIS reflectance data have been used in regional land-cover mapping, change detection, evapotranspiration estimation, and crop biomass estimation [41]–[44]. Based on the STARFM algorithm, some modified algorithms, such as enhanced STARFM (ESTARFM) [45] and Spatial Temporal Adaptive Algorithm for mapping Reflectance Change (STAARCH) [46], were proposed to improve the fusion performance. Based on the similar hypothesis with STARFM algorithm, Vanhellemont *et al.* [16] proposed a simple but effective method to fuse SEVIRI and MODIS marine reflectance data. The spatial and temporal fusion algorithms are for merging high temporal resolution, low spatial resolution images with low temporal resolution, and high spatial resolution images. The fusions have both high spatial resolution and high temporal resolution. Spectrally, only the spectral bands that have similar wavelengths of different sensors were fused. That is, the spatial–temporal fusion algorithms lose some spectral information.
3) Spatial, spectral, and temporal fusion algorithms, such as the unmixing-based fusion (UBF) algorithm [47]–[49], and some other algorithms based on UBF, such as the regularized UBF algorithm [50], [51] and the improved UBF algorithm [52]. For these algorithms, the spatial information is offered by TM-like images (or the TM-like-based classification image), while the spectral and temporal information is offered by MERIS images. These algorithms merge all useful information contained by multisource images. However, an essential assumption of those algorithms is that the temporal changing of a high spatial resolution image is negligible, which is unacceptable for rapidly changing inland water. In conclusion, there is not a proper image fusion algorithm for rapidly changing, optically complex inland water images. What is more, the three types of image fusion algorithms cannot cover all applications. For example, when fusing GOCI and Himawari images, both the spatial and spectral information are offered by the low temporal resolution images (GOCI). The current fusion algorithms cannot deal with that fusion well.

Aside from the theoretical limitations, a practical problem of applying the image fusion algorithms in water remote sensing is the cloud-contamination in the input images. Cloud-contaminated pixels cannot be used for water-color retrieval [53]. For image-fusion algorithms, cloud pixels are always masked before the fusion process. Therefore, the cloud pixels not only decrease the spatial–temporal coverage of the image; they also limit the fusion input image selection: for a specific study area, it is difficult to find multisource images that synchronize without any cloud covering. That problem undoubtedly limits the practicality of image fusion techniques in water color remote sensing. Thus, for rapidly changing inland water images, dealing with cloud-contaminated pixels is essential for fusion algorithms.

This research proposes a multitemporal image fusion (MTIF) algorithm based on a spectral and temporal unmixing method in two different domains (spatial–spectral and spatial–temporal). The algorithm belongs to the third type of image fusion algorithm, i.e., the algorithms that have high spatial, spectral, and temporal resolution. GOCI and Himawari images are fused to yield a product that has 500-m spatial resolution, eight spectral bands (in the visible–near-infrared spectral range), and 10-min temporal resolution. Furthermore, the temporal unmixing gave the fusion algorithm some outstanding characteristics (the undercloud information could be partly estimated) and great potential in water color remote sensing.

II. Data and Methods

A. Introduction to the MTIF Algorithm

A remote sensing image series can be seen as the dispersed sampling of the changing landscape's radiance information in three aspects: spatially (x- and y-dimensions), spectrally (z-dimension), and temporally (t-dimension). Different sensors capture the radiance information with different resolution in the three aspects. The general objective of an image fusion algorithm is to yield a unified, enhanced data set from images of different resolution in x-, y-, z-, and t-dimensions. Unifying different-resolution images in three aspects makes the fusion so complicated that many algorithms only focus on two aspects (as mentioned in Section I). In those algorithms, the unmixing based algorithms show great potential in fusing multisource images.

As a classical method to analyze remote sensing images, spectral unmixing separates a multispectral/hyperspectral image into abundance images and endmember spectra. In some studies, the abundance images and endmember spectra decomposed from multisource images show well consistency [54], [55]. Therefore, spectral unmixing offers an effective way to link multisource remote sensing images. The MTIF algorithm is based on spectral and temporal unmixing.

If a multispectral pixel with z-bands is composed of m endmembers, the mixing model can be expressed as

$$\mathbf{R} = \mathbf{C}\mathbf{r} + \boldsymbol{\mu} \tag{1}$$

where $\mathbf{R}$ is a $1 \times z$ matrix that represents the spectrum of the current pixel. $\mathbf{C}$ is a $1 \times m$ matrix that represents the abundance information. $\mathbf{r}$ is an $m \times z$ matrix that represents the endmember spectra. $\boldsymbol{\mu}$ is a $1 \times z$ matrix that represents the residual.

The traditional spectral unmixing was processed in the xyz domain [see Fig. 1(a)], the endmembers are the representative spectra in the scene, and the spatial abundance in each pixel represents the area percentage of different endmembers. The abundance image represents the spatial distribution of the endmembers. The endmembers were usually selected before unmixing using Pure Pixel Index (PPI) [56], N-Findr [57], and Vertex Component Analysis methods [58]. Then, the abundance of each endmember at each pixel could be calculated

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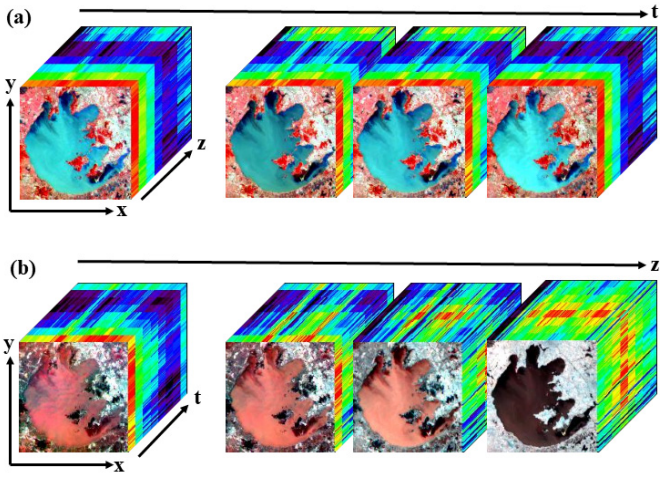


Fig. 1. Sketch map of (a) *xyz* and (b) *xyt* domains.

by minimization of the cost function as follows:

$$J_c = \sum_{\lambda} \left[R_{s,l,\lambda} - \sum_{m=1}^{M} \overline{C_{s,l,m}} \times r_{\lambda,m} \right]^2 \tag{2}$$

where $R_{s,l,\lambda}$ is the reflectance at wavelength λ of pixel (s,l), s and l represent the sample and line indexes, respectively; $r_{\lambda,m}$ is the reflectance of the mth endmember at wavelength λ; M stands for the total endmembers; and $\overline{C_{s,l,m}}$ is the estimated abundance of the mth endmember of pixel (s,l). Similarly, once the abundance information is known, the endmembers can also be estimated by minimizing the following cost function, mentioned by Zhukov *et al.* [47]:

$$J_r = \sum_{g=1}^{G} \sum_{s=1}^{S} \sum_{l=1}^{L} \left[R_{s,l,\lambda,g} - \sum_{m=1}^{M} C_{s,l,m,g} \times \overline{r_{\lambda,m}} \right]^2 \tag{3}$$

where $\mathbf{C}_{s,l,m,g}$ is the abundance of the mth endmember of pixel (s,l) in the gth input image; $\overline{\mathbf{r}_{\lambda,m}}$ is the estimated reflectance of the mth endmember at wavelength λ; S and L are the total samples and lines of the current image, respectively; G is the number of input images. Different from the estimation of abundance, the estimation of endmembers was processed band by band, not pixel by pixel. For an unmixing-based image fusion algorithm, its core mission is to find high spatial resolution $\bar{\mathbf{C}}$ and high spectral resolution $\bar{\mathbf{r}}$. Then the fusion reflectance $\bar{\mathbf{R}}$ can be determined from the following equation:

$$\bar{\mathbf{R}} = \bar{\mathbf{C}}\bar{\mathbf{r}}. \tag{4}$$

When the unmixing is expanded to the *xyt* domain [see Fig. 1(b)], the spectrum of a specific pixel represents the changing of remote sensed information at different times. The endmembers are representative temporal changing patterns in the scene, and the temporal abundance represents the pattern's spatial distribution. The composing and decomposing processes can also be described by (1)–(4). There are two assumptions in MTIF algorithm.

1) The abundance is not sensitive to the spectrum resolution. For one specific pixel. When the spatial resolution of multisource images is the same, the abundance calculated from different spectral-resolution images should be equal to each other.
2) The endmembers are not sensitive to the spatial and temporal resolution. For a specific area that is covered by quasisynchronous multisource remote sensing images, when the multisource images have the same band settings but different spatial and temporal resolution. The endmembers extracted from different sensors should be equal to each other.

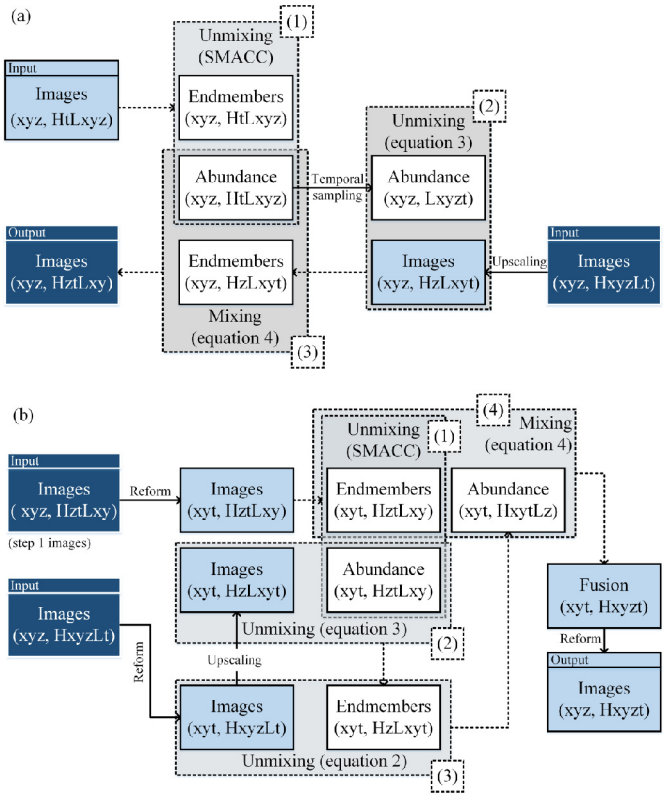


Fig. 2. Flowchart of (a) step 1 and (b) step 2 of MTIF.

The MTIF algorithm contains two steps. Step 1 is processed in the *xyz*-domain [see Fig. 2(a)]. The yielded images have high spectral and temporal resolution but low spatial resolution. Based on the step 1 yielded images, step 2 is handled in the *xyt* domain [see Fig. 2(b)]. The resulting images will have high spatial, spectral, and temporal resolution. The detailed steps of the algorithm are shown in Section II-B.

B. Algorithm Steps

In the flowchart, Hxy, Hz, and Ht represent relatively high spatial resolution, spectral resolution, and temporal resolution, respectively. Lxy, Lz, and Lt represent relatively low spatial resolution, spectral resolution, and temporal resolution, respectively. This research is focused on the fusion of GOCI and Himawari images, which are HxyzLt images and HtLxyz images, respectively.

In step 1, the input images are HtLxyz images and HxyzLt images in the *xyz* domain. The final goal of this step is to generate HztLxy images, which are the input of the next step

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[see Fig. 2(b)]. In detail, the fusion contains three substeps [those substeps are marked in Fig. 2(a) by (1)–(3)].

1) The input HtLxyz images are decomposed HtLxyz endmembers and HtLxyz abundance images using the sequential maximum angle convex cone (SMACC) algorithm (the unmixing algorithm is flexible).
2) The HtLxyz abundance images at the HxyzLt images' acquisition time were picked to be the Lxyzt abundance images. According to assumption (1), the Lxyzt abundance images equals the HzLxyz abundance images. Then, based on all the Lxyzt abundance and upscaled HxyzLt images (i.e., HzLxyt images), the HzLxyt endmembers are calculated using (3). In this article, the upscaling was processed by averaging the high spatial resolution pixels in each low spatial resolution pixel boundaries.
3) The HztLxy images are composed using the HtLxyz abundance [equal to the HztLxy abundance, according to assumption (1)] images and HzLxyt endmembers [equals the HztLxy endmembers, according to assumption (2)] using (4).

In step 2, the input images are the step 1 yielded HztLxy images and the original HxyzLt images in the *xyz* domain. The goal of this step is to generate Hxyzt images. The HztLxy and HxyzLt images are first transposed from the *xyz* domain to the *xyt* domain. An image in the *xyt* domain is the time series of a specific spectral band [see Fig. 1(b)]. Then, the fusion is processed for each spectral band, separately. The fusion of low temporal resolution image series and high temporal resolution image series could be seen as a multispectral–hyperspectral fusion problem. There are some classic algorithms to solve that problem, such as principal component analysis (PAC) [26], nearest neighbor diffusion (NND) [59], GS [26], and UBF [47]. In the MTIF algorithm, the problem is solved based on the spectral unmixing theory, as in Step 1, and the advantages of using this theory are described in Section III-B4. The fusion of each spectral band is processed in four substeps [those substeps are marked in Fig. 2(b) by (1)–(4)].

1) The HztLxy images are decomposed to HztLxy endmembers and HztLxy abundance images using the SMACC algorithm.
2) Based on the HztLxy abundance [equal to the HzLxyt abundance, according to assumption (1)] and the upscaled HxyzLt images (i.e., HzLxyt images), the HzLxyt endmembers are calculated using (3).
3) The HxyzLt abundance images are calculated based on the HxyzLt images and HzLxyt endmembers [equal to the HxyzLt endmembers, according to assumption (2)] using (2).
4) The final fusion images are composed using HztLxy endmembers [equal to the Hxyzt endmembers, according to assumption (2)] and HxyzLt abundance [equal to the Hxyzt abundance, according to assumption (1)] images using (4) and then transposed back to the *xyz* domain images.

The fusion is done when input images of all the spectral bands are processed.

In step 2 of the MTIF algorithm, the unmixing in the *xyt* domain is technically the same with that in the *xyz* domain except for a special situation: when the pixel was contaminated by cloud. Cloud coverage of one specific pixel in the *xyt* domain could be separated into two situations: 1) there is cloud cover for much of the period (when the number of clear band is less than the number of endmember) and 2) there is cloud cover for part of the period (specifically, the clear band number is larger than the endmember number). In the first situation, the images cannot provide enough information to the MTIF algorithm, thus the fusion cannot be processed, and the current pixel is marked as cloudy. For the second situation, if the clean pixel band number is greater than the endmember number (to make sure that the unmixing system is positive-definite), the MTIF algorithm could recover the information under the cloud: The cloud-contaminated bands were first assigned 0 in each *xyt* image. Then, in the HtzLxy image endmember calculation step [step 2, substep (2)], the endmember in the clouded image is yielded using the clean pixels using (3). In the Hxyzt abundance calculation step [step 2, substep (3)] because the cloud-covered pixels were assigned 0, the cloud-covered pixels would not make any contribution to the linear system [see (2)], which means that only the clean band's temporal pattern is used to generate the abundance of the pixel. The theoretical hypothesis is that the clean band's pixels could reflect most of the temporally changing information. With the HtzLxy endmember and HxzyLt abundance, the final fusion with estimated undercloud information is yielded.

C. Image Data and Its Processing

GOCI was the first ocean color sensor launched in a geostationary orbit. It has a revisit time of approximately 1 h and can readily monitor the movement of water [60]. What is more, its COMS sensor has a high signal-to-noise ratio and 500-m spatial resolution, which makes GOCI images widely used in water color remote sensing research [61], [62]. GOCI level-1b production images were used in this article. The images were downloaded from the Korea Ocean Satellite Center (http://kosc.kiost. ac/eng/).

Himawari-8 was launched by the Japan Meteorological Agency (JMA) on October 7, 2014, and it started running on July 7, 2015. It carries a Himawari imaging system [Advanced Himawari Imager (AHI)] with a high spectral, temporal, and radiometric resolution. AHI has 16 visible and infrared channels. The spatial resolution is 0.5–2 km, and the temporal resolution is 10 min [63], [64]. In this article, we used Himawari-8 Level 1 full-disk gridded data. The data can be obtained from EORC (http://www.eorc.jaxa.jp/ptree). For the 0.5-/1-km Himawari Standard Data (HSD), only the images from the last 30 days are provided by the JMA. Thus, the Himawari L1 gridded data, with spatial resolution in 400–1000-nm bands of 2 km, were used as the algorithm input data set because it could offer image data from July 2015.

The spectrum response functions of GOCI and Himawari are shown in Fig. 3. Compared with Himawari, GOCI has more thin bands. This means that GOCI could capture more water optical properties.

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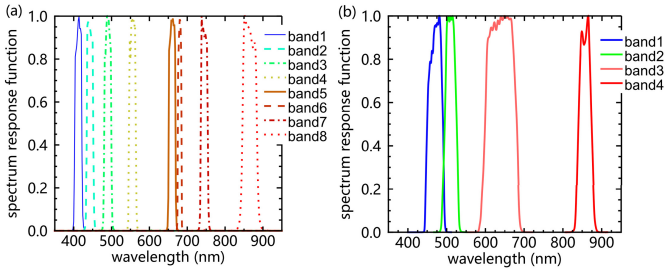


Fig. 3. Spectrum response functions of (a) GOCI and (b) Himawari within 400–900 nm.

Geometric correction, the unitless top-of-atmosphere reflectance (R_{TOA}) calculation, and atmospheric correction were carried out for Himawari and GOCI images. To avoid the effect of atmospheric correction errors for the fusion algorithms, R_{TOA} images were fused, and then atmospherically corrected to calculate C_{chla}.

The geometric correction of both Himawari L1 and GOCI L1b data were processed using the geographical lookup table (GLT). Then, the Himawari images were resampled to the GOCI's grid using the nearest-neighbor method.

GOCI L1b and Himawari L1 data provide radiance and albedo information, respectively, from the top of the atmosphere. Therefore, the next step is to transform the radiance and albedo into reflectance at the top of the atmosphere. For GOCI L1b images, the following equation is used [65], [66]:

$$R_{\mathrm{TOA}}(\lambda) = \frac{\pi L_{\mathrm{TOA}}(\lambda)}{F_0(\lambda)\cos(\theta)} \tag{5}$$

where L_{TOA} is the measured radiance in mW/msr; F_0 is the extraterrestrial solar irradiance corrected for each day of the year measured in mW/m; λ is the wavelength; and θ refers to the solar zenith angle in degrees for each pixel, which for GOCI is a standard band in the L1b data set itself.

For Himawari images, the following equation is used:

$$R_{\mathrm{TOA}}(\lambda) = \frac{\mathrm{albedo}(\lambda)}{\cos(\theta)}. \tag{6}$$

The atmospheric correction was carried out using the UAVC algorithm (atmospheric correction based on ultraviolet wavelength) established by He *et al.* [67].

D. Fusion Quality Evaluation

The fusions were evaluated both qualitatively and quantitatively. The quantitative assessment was performed using several distortion metrics between the input images and the upscaled fusion images. We first selected Pearson's correlation coefficient (cc), to evaluate the coherence between the fusion and the original images, which could be calculated by (7), as shown at the bottom of this page, where cc_λ is the cc of band λ; $R_{\mathrm{TOA}}^{\mathrm{true}}$ and $R_{\mathrm{TOA}}^{\mathrm{fused}}$ are the R_{TOA} of the reference image and the fused image, respectively; $\overline{R_{\mathrm{TOA}}^{\mathrm{true}}}$ and $\overline{R_{\mathrm{TOA}}^{\mathrm{fused}}}$ are the average of $R_{\mathrm{TOA}}^{\mathrm{true}}$ and $R_{\mathrm{TOA}}^{\mathrm{fused}}$, respectively. The final cc index of each image pair is the average cc_λ value of all available wavelength.

Then, indices that are more appropriate to evaluate multispectral images, such as the spectral angle (SA) and the relative dimensionless global error (ERGAS) index, were also computed as

$$\mathrm{SA} = \frac{1}{S\times L}\sum_{s=1}^{S}\sum_{l=1}^{L}\cos^{-1}\left[\frac{\sum_{\lambda} R_{\mathrm{TOA}}^{\mathrm{true}}(s,l,\lambda)\times R_{\mathrm{TOA}}^{\mathrm{fused}}(s,l,\lambda)}{\sqrt{\sum_{\lambda} R_{\mathrm{TOA}}^{\mathrm{true}}(s,l,\lambda)^2}\times\sqrt{\sum_{\lambda} R_{\mathrm{TOA}}^{\mathrm{fused}}(s,l,\lambda)^2}}\right] \tag{8}$$

$$\mathrm{ERGAS}_{\mathrm{scale}} = 100\frac{S_h}{S_l}\sqrt{\frac{1}{z}\sum_{\lambda}\left(\mathrm{RMSE}_\lambda^2/M_\lambda^2\right)}$$

$$\mathrm{RMSE}_\lambda = \frac{1}{S\times L}\sqrt{\sum_{s=1}^{S}\sum_{l=1}^{L}\left(\mathbf{R}_{\mathrm{TOA}}^{\mathrm{true}}(s,l,\lambda)-\mathbf{R}_{\mathrm{TOA}}^{\mathrm{fused}}(s,l,\lambda)\right)^2} \tag{9}$$

where S_h and S_h are the high and low spatial resolution, respectively. In our experiment, S_h and S_l are 500 and 2000, respectively; S and L are the total samples and lines of the current image; M_λ is the average value of band λ.

Finally, to evaluate the relative errors of the fusions, the mean absolute percentage error (MAPE) index was calculated by

$$\mathrm{MAPE}_\lambda = \frac{1}{S\times L}\sum_{s=1}^{S}\sum_{l=1}^{L}\left|\frac{R_{\mathrm{TOA}}^{\mathrm{true}}(s,l,\lambda)-R_{\mathrm{TOA}}^{\mathrm{fused}}(s,l,\lambda)}{R_{\mathrm{TOA}}^{\mathrm{true}}(s,l,\lambda)}\right|. \tag{10}$$

The final MAPE index of each image pair is the average MAPE_λ of all available wavelength.

Considering the band setting difference between Himawari and GOCI, the three indexes were calculated only on the fusions and GOCI images. In practice, the ideal fusions at the Hxyz images' acquisition time should be GOCI images. In this article, only eight fusions could be evaluated by the reference images. In principle, the reference images could not be the algorithm's input images at the same time. Therefore, to better assess the fusion performance, the fusion was processed for eight times. In each fusion, 1 GOCI image was selected as the reference image, while the left GOCI images were selected as the MTIF input images to yield the fusions. The fusion

$$\mathrm{cc}_\lambda = \frac{\sum_{s=1}^{S}\sum_{l=1}^{L}\left(R_{\mathrm{TOA}}^{\mathrm{true}}(s,l,\lambda)-\overline{R_{\mathrm{TOA}}^{\mathrm{true}}(\lambda)}\right)\left(R_{\mathrm{TOA}}^{\mathrm{fused}}(s,l,\lambda)-\overline{R_{\mathrm{TOA}}^{\mathrm{fused}}(\lambda)}\right)}{\sqrt{\sum_{s=1}^{S}\sum_{l=1}^{L}\left(R_{\mathrm{TOA}}^{\mathrm{true}}(s,l,\lambda)-\overline{R_{\mathrm{TOA}}^{\mathrm{true}}(\lambda)}\right)^2}\sqrt{\sum_{s=1}^{S}\sum_{l=1}^{L}\left(R_{\mathrm{TOA}}^{\mathrm{fused}}(s,l,\lambda)-\overline{R_{\mathrm{TOA}}^{\mathrm{fused}}(\lambda)}\right)^2}} \tag{7}$$

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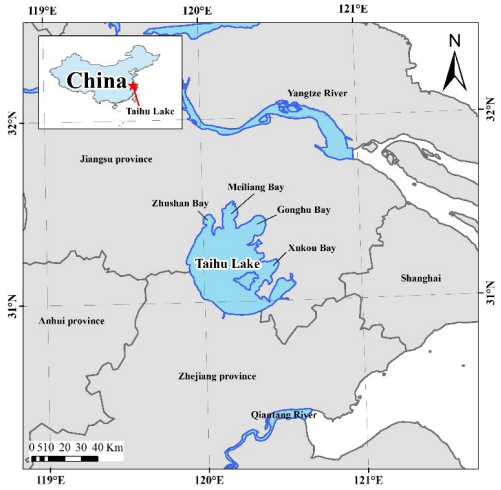


Fig. 4. Location of the study area.

image at the reference time was then picked to compare with the reference image using the assessment indexes. After eight fusions were done, the average indexes were calculated to be the final performance of each algorithm.

Theoretically, the fusing of GOCI and Himawari images is not supported by current fusion algorithms. However, in step 2 of the MTIF algorithm, the fusion of reformed Hzt and Hxyz images in the temporal domain could be seen as a multihyperspectral fusion process. Therefore, four algorithms (GS, NND, PC, and UBF) were used to fuse the MTIF Step 1 result and GOCI image in the temporal domain. The four algorithms were applied based on the MTIF Step 1 fusions, so, in all the following sections, those algorithms are expressed as s1GS, s1NND, s1PC, and s1UBF, respectively.

E. C_{chla} Estimation Model

The concentration of chlorophyll-a (C_{chla}) is a key indicator of biophysical status, water quality, and pollution in inland water environments. C_{chla} can be estimated from remotely sensed data with wide spatial and temporal coverage because of its unique optical properties. Huang *et al.* [12] proposed a simple regional NIR-red two-band empirical algorithm of C_{chla} for GOCI for Taihu Lake. Their use of a large *in situ* data set ($N = 1126$) showed that the algorithm exhibits reliable performance in the study area $R^2 = 0.71$, $p < 0.001$). Thus, we use that algorithm to estimate water C_{chla} in Taihu Lake using the fusion images. The regression equation is

$$\begin{aligned} C_{\text{chla}} &= 10^{F(\text{ratio})} \\ F(\text{ratio}) &= 1.8875 + 0.8296 * (1 - 0.2241^{\log(\text{Rrs}(745)/\text{Rrs}(680),10)}) \end{aligned} \tag{11}$$

where Rrs(745) and Rrs(680) are the atmospherically corrected reflectances of the 680- and 745-nm bands of the GOCI and fused images, respectively.

III. Experiment

A. Experimental Setup

The study area, Taihu Lake, located between the Jiangsu and Zhejiang Provinces of China (see Fig. 4), is a large shallow lake with an area of ~2338 km^2 and a mean depth of ~1.9 m [68]. The lake suffers from water quality deterioration and also shows high turbidity in some areas due to a large quantity of sediment resuspension. These factors have led to the presence of complex water conditions. High frequency and precise monitoring of Taihu Lake is urgently needed.

Two data sets were chosen to validate the MTIF method in this article: 1) 8 GOCI images (from 8:30 to 15:30, 1-h interval) and 51 Himawari images (from 8:00 to 16:50, 10-min interval, data at 8:10, 10:10, and 10:40 were missing) from February 22, 2016. The image data were cloud-free and were used to evaluate the visual effect, quantitative accuracy, and chlorophyll-a concentration estimation of the MTIF algorithm and 2) 8 GOCI images (from 8:30 to 15:30, 1-h interval) and 52 Himawari images (from 8:00 to 16:50, 10-min interval, data at 8:30 and 10:40 were missing) from July 22, 2016. These images were partly cloudy and optically complex (an algae bloom was taking place and changing fast temporally in Zhushan Bay). That data set was used to evaluate the undercloud information estimation effect, and the fast-changing water property capturing ability of the MTIF algorithm.

To evaluate the undercloud reflectance estimation ability of the MTIF algorithm, pieces of Himawri images (40×40 pixels) were cut from the original images as the experimental Himawari data set. Meanwhile, the GOCI images of the same area were also extracted. Then, in the experimental Himawari data set, randomly selected images were marked as the cloudy scene. In those scenes, half of the image (20×40 pixels) was marked as cloud pixels. In the fusion process, the image reflectance under the simulated cloud was compared with the true reflectance to evaluate the undercloud reflectance estimation accuracy.

The C_{chla} maps were produced by using the fusions and GOCI images to evaluate the C_{chla} estimation potential of the MTIF algorithm.

B. Data Fusion Quality

1) Visual Effect: Himawari, GOCI, and their fusions at 12:30 on February 22, 2016, are shown in Fig. 5. Visually, the color of MTIF [see Fig. 5(d)] and s1NND [see Fig. 5(g)] fused images are both close to the target GOCI image [see Fig. 5(a)] in color. However, the spatial detail information of s1NND fusion [see Fig. 5(g)] was blurred, especially in the northwest region of the lake. Compared with the s1NND fusion [see Fig. 5(g)], the s1GS [see Fig. 5(e)], s1UBF [see Fig. 5(f)], and s1PC [see Fig. 5(h)] fusions perform better. Among those three fusion algorithms, the UBF algorithm recovered most of the spatial detail information, but some outlier points occurred around water-land boundaries. One possible reason for those outlier points is that, in the UBF algorithm, a fixed number of classes was set (20 classes in this article). Considering that water pixels are much darker than land pixels, a large class number is necessary to obtain the water's spatial detail information. Therefore, the sliding window overlapping the water-land boundaries contains a lot of classes. The increasing variables made the unmixing process unstable. That problem was also shown in Zhukov's research [47]. The s1GS

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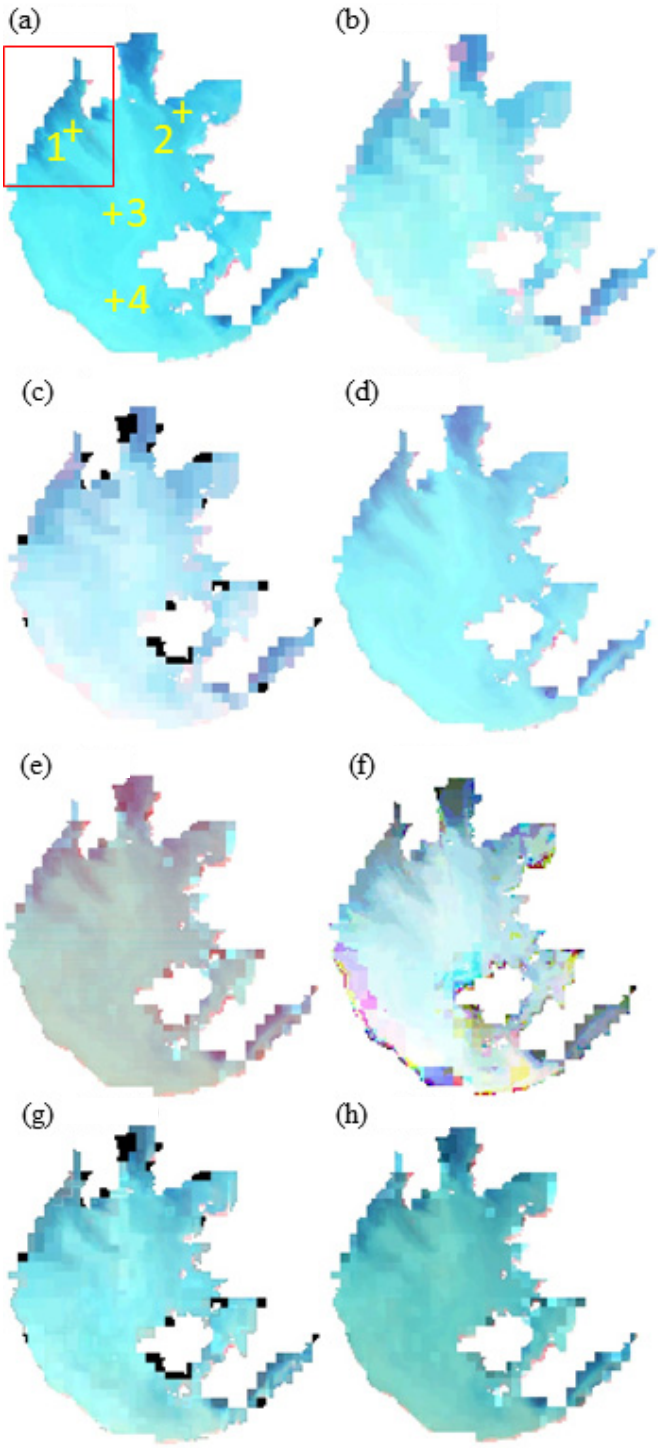


Fig. 5. Maps of (a) GOCI image (R: band 7, G: band 4, B: band 3), (b) Himawari image (R: band 4, G: band 3, B: band 2), (c) s1 image (R: band 7, G: band 4, B: band 3), (d) MTIF fusion (R: band 7, G: band 4, B: band 3), (e) s1GS fusion (R: band 7, G: band 4, B: band 3), (f) s1UBF fusion (R: band 7, G: band 4, B: band 3), (g) s1NND fusion (R: band 7, G: band 4, B: band 3), and (h) s1PC fusion (R: band 7, G: band 4, B: band 3) of Taihu Lake at 12:30 on February 22, 2016.

[see Fig. 5(e)] and s1PC [see Fig. 5(h)] fused images could capture part of the spatial detail, but low-resolution boundaries could be observed in those images. What is more, the image color looked darker than the target image. The detailed images

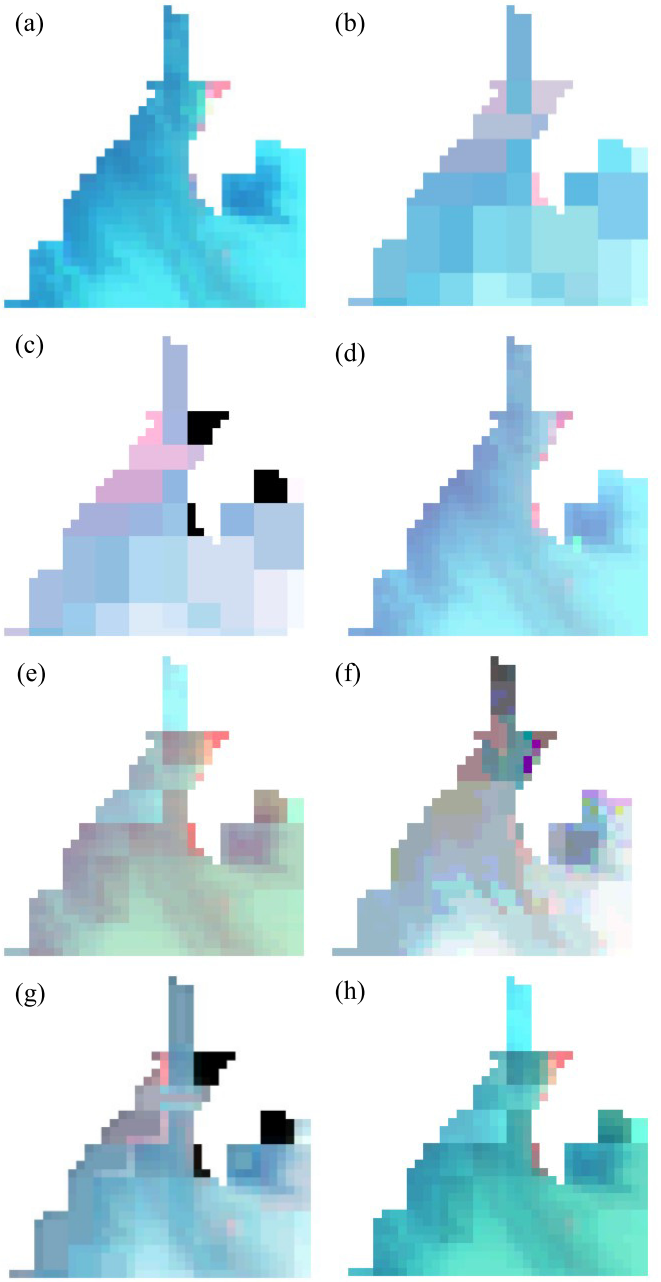


Fig. 6. Maps of (a) GOCI image (R: band 7, G: band 4, B: band 3), (b) Himawari image (R: band 4, G: band 3, B: band 2), (c) s1 image (R: band 7, G: band 4, B: band 3), (d) MTIF fusion (R: band 7, G: band 4, B: band 3), (e) s1GS fusion (R: band 7, G: band 4, B: band 3), (f) s1UBF fusion (R: band 7, G: band 4, B: band 3), (g) s1NND fusion (R: band 7, G: band 4, B: band 3), and (h) s1PC fusion (R: band 7, G: band 4, B: band 3) of the red rectangle marked region in Fig. 5.

in the red rectangle marked region in Fig. 5(a) (see Fig. 6) show another advantage of the UBF algorithms (i.e., MTIF and UBF in this article): affected by the land pixels, some outlier pixels can be observed in s1 fusions [the black regions in Zhushan Bay in Fig. 6(c)]. The fusions of s1NND [see Fig. 6(g)], s1GS [see Fig. 6(e)], and s1PC [see Fig. 6(h)] partly kept those dark pixels. However, by including the unmixing process, the endmembers' spectra were calculated using all pixels in the scene (or in the sliding window). MTIF and UBF algorithms controlled the random errors better, to generate

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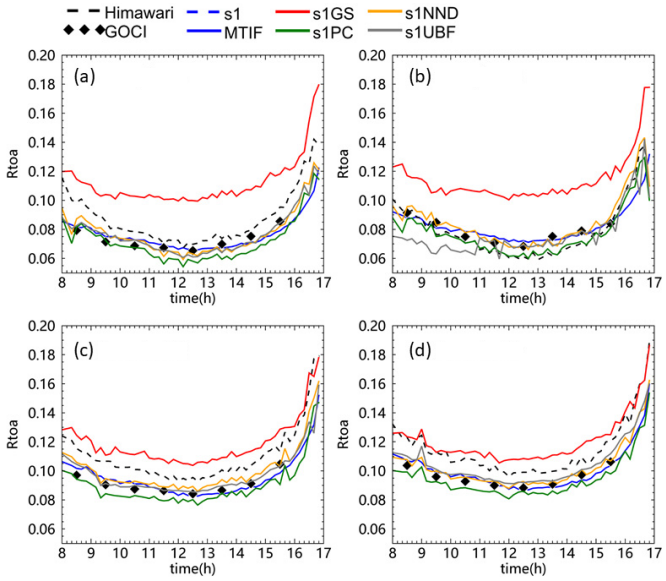


Fig. 7. Temporal changing of band 8 R_{toa} of GOCI and the different fusions and band 4 R_{toa} of Himawari at (a) point 1, (b) point 2, (c) point 3, and (d) point 4.

clearer spatial detail. In conclusion, the MTIF fused images have the best visual effect.

Four representative sample points in Taihu Lake (marked by yellow crosses in Fig. 5) were selected to show the temporal changing of Himawari, GOCI, and the fused images (see Fig. 7). Because only band 4 of Himawari and band 8 of GOCI have similar band settings (the same center wavelengths), only those two bands were compared in Fig. 7. The results show that, first, GOCI and Himawari images have similar temporally changing patterns: R_{toa} decreased before 12:30 in all four sampling stations, then increased until 17:00. Second, s1 fusions are closer to GOCI R_{toa} lines instead of Himawari's. That means the spectral information was successfully fused into Ht images in MTIF's Step 1 process. Finally, the Himawari R_{toa} of band 4 is relatively higher than those of GOCI band 8. An ideal fusion result should be similar to GOCI images numerically and similar to Himawari images in temporal details. For the six fusion methods, s1GS fusions are generally higher than the reference line. In contrast, the s1PC fusions are systematically lower than GOCI images. The MTIF, s1NND, and s1UBF had better performance in recovering the high temporal R_{toa} information.

2) Quantitative Assessment: The ERGAS, cc, SA, and MAPE indexes of the MTIF algorithm are compared with those of the MTIF s1, s1GS, s1NND, s1PC, and s1UBF (see Table I). The MTIF algorithm's performance is the best for all evaluation indexes. For cc, SA, and MAPE indexes, GS, NND, PC, and UBF fusions have similar, or even worse, performance compared with Step 1 fusions (SA, especially). That means the proposed algorithm could effectively fuse high spatial and temporal resolution information.

3) C_{chla} Estimation Ability Validation: C_{chla} of Taihu Lake on February 15, 2016, was calculated using MTIF Step 1, and final fusions, s1UBF, s1NND, s1GS, and s1PC fusions were compared with those using GOCI images. The results at 12:30 are shown in Fig. 8. Generally, chlorophyll-a concentration was higher in the southern part of the lake. The s1 C_{chla} maps [see Fig. 8(b)] reflect the general spatial distribution well. However, affected by land pixels, some high C_{chla} points

TABLE I

QUALITY ASSESSMENT USING SEVERAL METRICS BETWEEN THE FUSIONS YIELDED FROM DIFFERENT METHODS AND GOCI IMAGES

	ERGAS	cc	SA	MAPE
MTIF	**1.1463**	**0.8423**	**1.0019**	**0.0277**
s1	4.3612	0.2232	1.5391	0.0689
s1GS	1.4063	0.4867	1.0317	0.0289
s1NND	3.9914	0.2629	1.2874	0.0614
s1PC	1.6056	0.5397	1.0386	0.0289
s1UBF	4.2413	0.2540	3.9797	0.0809

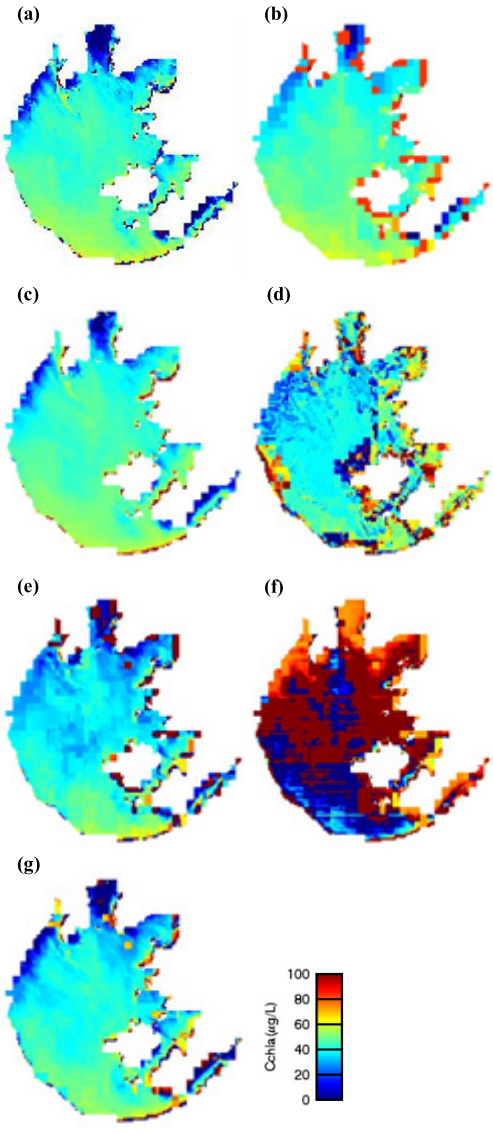


Fig. 8. C_{chla} distribution of Taihu Lake yielded from (a) GOCI images, (b) s1 images, (c) MTIF images, (d) s1UBF images, (e) s1NND images, (f) s1GS images, and (g) s1PC images at 12:30 on February 15, 2016.

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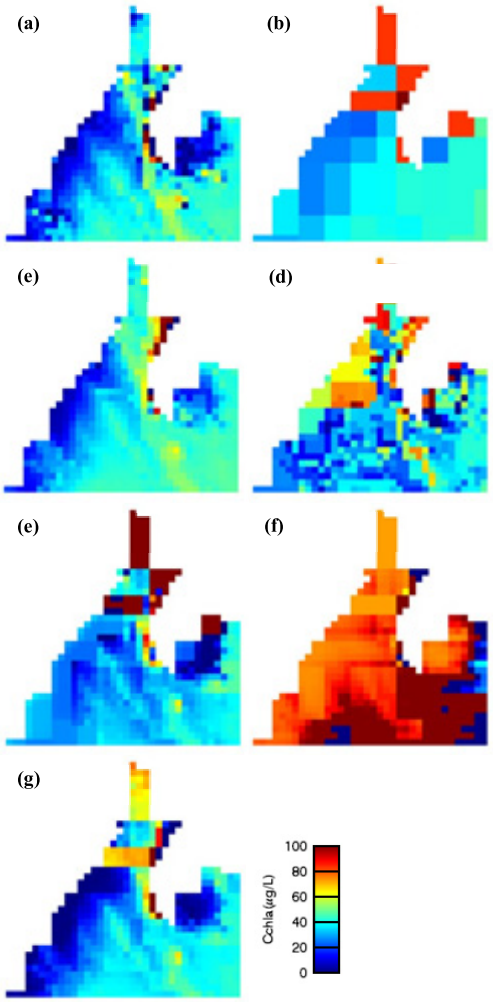


Fig. 9. C_{chla} distribution of Taihu Lake yielded from (a) GOCI images, (b) s1 images, (c) MTIF images, (d) s1UBF images, (e) s1NND images, (f) s1GS images, and (g) s1PC images of the marked area in Fig 5(a).

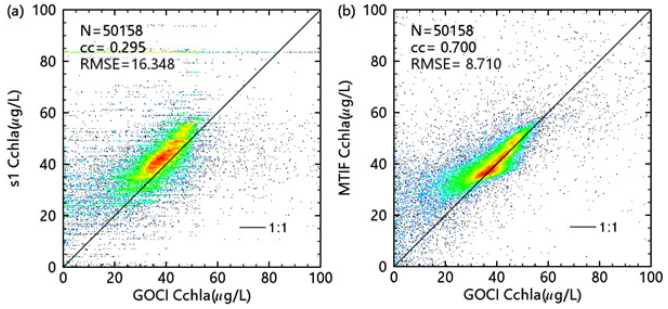


Fig. 10. Scatter plots of comparison of the C_{chla} estimated from GOCI with those from (a) s1 and (b) MTIF fusions in Taihu Lake on February 15, 2016.

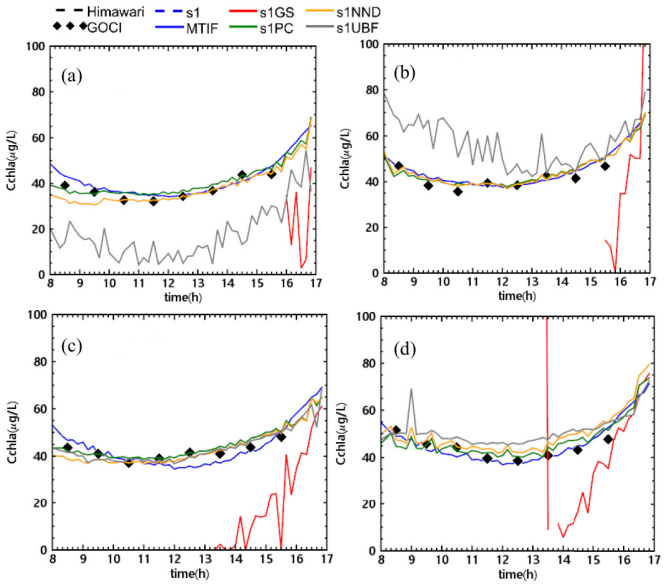


Fig. 11. Plots of C_{chla} trends yielded from Himawari, GOCI, and MTIF fused images of (a) sample 1, (b) sample 2, (c) sample 3, and (d) sample 4 in Fig. 5.

are distributed in near-shore regions. The MTIF C_{chla} maps [see Fig. 8(c)] successfully wiped out the outlier high-value points. The s1PC algorithms also performed well in C_{chla} map generation [see Fig. 8(g)]. Aside from some low-resolution pixel boundaries, s1PC had similar performance with the MTIF algorithm. For the s1UBF and s1NND C_{chla} maps [see Fig. 8(d) and (e)], the high spatial resolution information was not fused well. As mentioned before, s1UBF fusions have a lot of outlier pixels on the water–land boundaries. Those errors were transferred into the final C_{chla} maps. In the s1NND C_{chla} maps, low spatial resolution pixels could be clearly observed. Affected by the spectral distortion, s1GS algorithm nearly failed in C_{chla} estimating. Another advantage of the MTIF algorithm is shown in Zhushan Bay region (see Fig. 9): in the GOCI C_{chla} map, some salt-and-pepper noise could be observed in the low-C_{chla} regions. Those errors are not obvious in RGB-composed images. However, the numerical calculation among different spectral bands enhanced the random errors. By introducing a spectral unmixing process, the MTIF yielded C_{chla} map controlled the salt-and-pepper noise well.

C_{chla} map yielded from GOCI image was also compared with those from s1 and MTIF fusions in scatter plots (see Fig. 10). The results show that the s1 fusions systematically underestimated C_{chla}. Affected by outlier points, the cc and RMSE indexes are not satisfied (cc = 0.295, RMSE = 16.348). On the other hand, the MTIF-yielded C_{chla} values have a much higher correlation and lower error with GOCI-yielded C_{chla} (cc = 0.700, RMSE = 8.710). What is more, by downscaling the s1 fusions, the underestimation was corrected well; the points are generally closer to the 1:1 line.

Then, the temporal trends of the C_{chla} from the GOCI images, MTIF, s1GS, s1PC, s1NND, and s1UBF fusions were compared at the four representative sample points (the four points in Fig. 5). The results (see Fig. 11) indicate that the MTIF, s1PC, and s1NND C_{chla} trends are stable and numerically coherent with the GOCI C_{chla} trends. By combining the GOCI images' spectral information, MTIF fusions successfully captured the C_{chla} information using better spectral settings while keeping the Himawari's temporal resolution. The s1UBF algorithm performed well at some points [see Fig. 11(c) and (d)]. However, in Fig. 11(a) and (b), the s1UBF lines show a significant offset compared with the GOCI C_{chla} lines. In conclusion, the s1-based fusions are all generated in the *xyt* domain, and the spectral information was fused band-by-band to keep the high spectral fidelity. Therefore, both the

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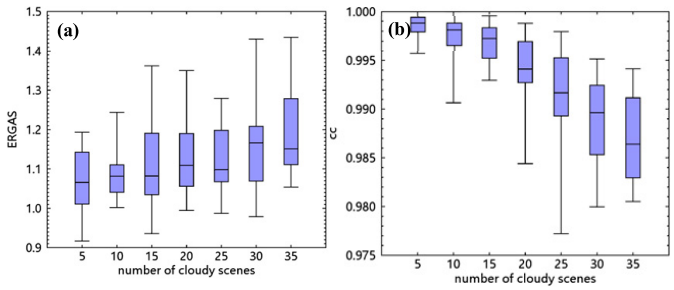


Fig. 12. Changing (a) ERGAS and (b) cc with an increasing number of cloudy scenes.

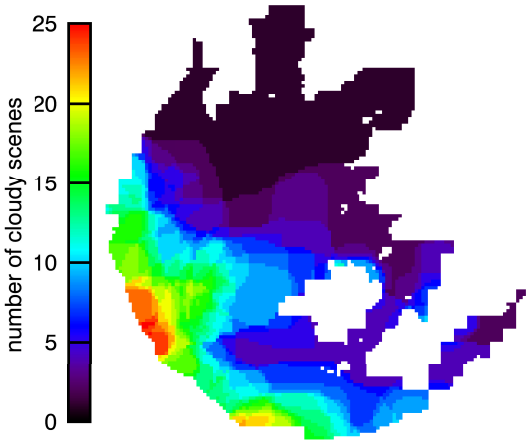


Fig. 13. Maps of cloudy pixels' spatial distribution over Taihu Lake 2016.

MTIF, s1PC, and s1NND algorithms could recover the high temporal resolution C_{chla} trends well.

4) Undercloud Information Estimation Ability Evaluation: The undercloud information estimation accuracy was evaluated using ERGAS and R indexes. 5, 10, 15, 20, 25, 30, and 35 scenes in total 52 clean Himawari scenes were randomly marked as cloudy images. Then GOCI and Himawari sample images were fused and compared with the reference images (the fusions yielded by the MTIF algorithm using 52 clean Himawari and 8 clean GOCI sample images). The cloudy image simulation and fusion in each cloudy situation was repeated 20 times to control the random error. The results show that, when the number of cloudy scenes is less than 30, the ERGAS index stays at a steady level (around 1.1). With the cloudy scenes further increasing, the ERGAS index reached around 1.15 [see Fig. 12(a)]. For cc index, the index values kept decreasing with number of cloudy scenes increasing; when the number of cloudy scenes reached 35, the average cc value was still high (0.986) [see Fig. 12(b)]. Under that situation, only 32.6% of the information was offered by the input images, which means that the temporal unmixing method in the proposed MTIF algorithm has a strong ability to predict undercloud reflectance information.

To further verify the undercloud information predicting ability, Himawari and GOCI images of Taihu Lake on July 22, 2016, were fused using the MTIF algorithm. The cloudiest areas were distributed in the southwest region of the lake (see Fig. 13). There were up to 25 cloudy Himawari scenes in that area. With GOCI's temporal resolution, GOCI and

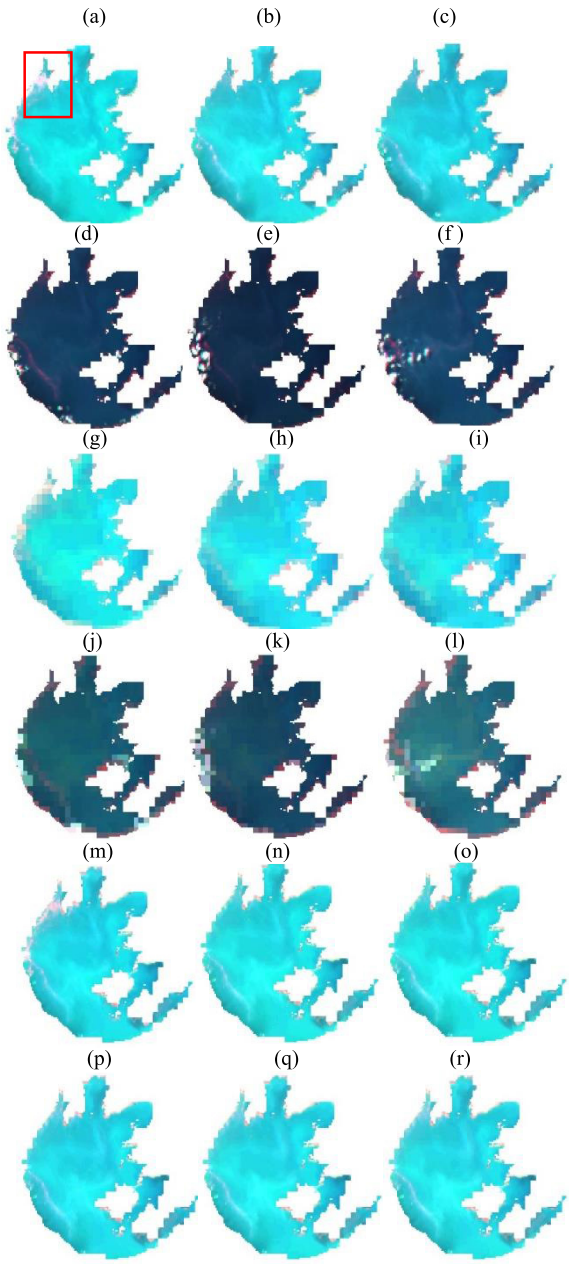


Fig. 14. GOCI images (R: band 7, G: band 4, B: band 3) at (a) 9:30, (b) 10:30, (c) 11:30, (d) 12:30, (e) 13:30, and (f) 14:30. Himawari images (R: band 4, G: band 3, B: band 2) at (g) 9:30, (h) 10:30, (i) 11:30, (j) 12:30, (k) 13:30, and (l) 14:30. MTIF fusions (R: band 7, G: band 4, B: band 3) at (m) 9:30, (n) 10:30, (o) 11:30, (p) 12:30, (q) 13:30, (r) and 14:30.

Himawari images between 9:30 and 14:30 [see Fig. 14(a)–(l)] showed that after 12:30, some cloud contaminated pixels appeared in the southwest region of the lake. The MTIF fusion image (the fifth and sixth line of Fig. 14) successfully estimated the information under the cloud shown in the GOCI and Himawari images. The striped algal bloom under cloud could be clearly observed.

IV. Discussion

A. Endmembers' Coherency

An important hypothesis in MTIF is that the spectral and temporal endmembers of Himawari images match those of

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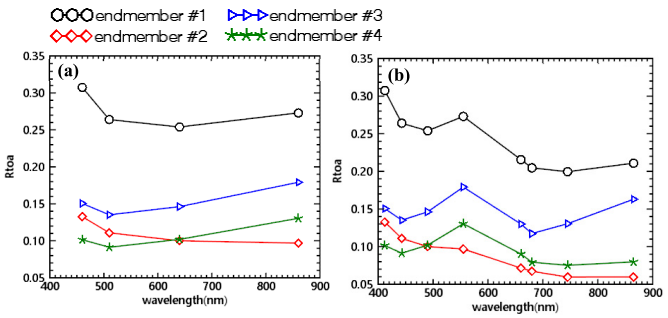


Fig. 15. Endmembers of (a) Himawari image and (b) GOCI image of Taihu Lake on February 15, 2016.

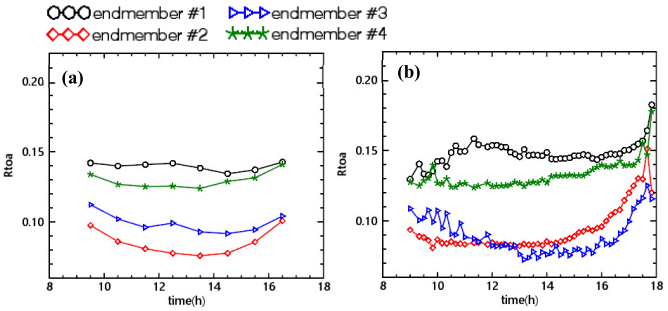


Fig. 16. Endmembers of (a) GOCI temporal image band 7 and (b) s1 fusion temporal image band 7 of Taihu Lake on February 15, 2016.

GOCI images. In the *xyz* domain, all of the image pixels were collected and unmixed by the SMACC algorithm. The endmember spectra of Himawari images of Taihu Lake on February 15, 2016, are shown in Fig. 15(a). Because Himawari images have four spectral bands between 400 and 900 nm, the endmember number was set to 4 in order to make the unmixing system positive-definite. The GOCI endmembers, inversed from GOCI images and Himawari abundance images, are shown in Fig. 15(b). The result indicates that, although numerical bias exists between the two sets of endmembers, the reflectance trends of the corresponding endmembers remained the same. In the *xyt* domain, the temporal endmembers of the GOCI and the step 1 fusion images' chlorophyll-a sensitive band—the 745-nm band (band 7)—are shown in Fig. 16. Similar to those in the *xyz* domain, corresponding temporal endmembers (plots with the same symbols) have similar trends. Step 1 endmembers showed more detailed temporal variation than the GOCI endmembers.

In conclusion, the endmember spectra calculated from (3) could keep the low spectral/temporal resolution endmembers' spectral trends while capturing the low spatial resolution images' varied spectral/temporal information.

B. Changing Monitoring Ability

In the marked region in Fig. 14, an algal bloom took place in the south of Zhushan Bay at 9:30. However, the bloom rapidly vanished by 10:30. That is, the 1-h temporal resolution of GOCI is not sufficient for the rapidly changing bio-physical processes in Taihu lake. From the 10-min resolution of Himawari false color composite images [see Fig. 17(c)–(h)],

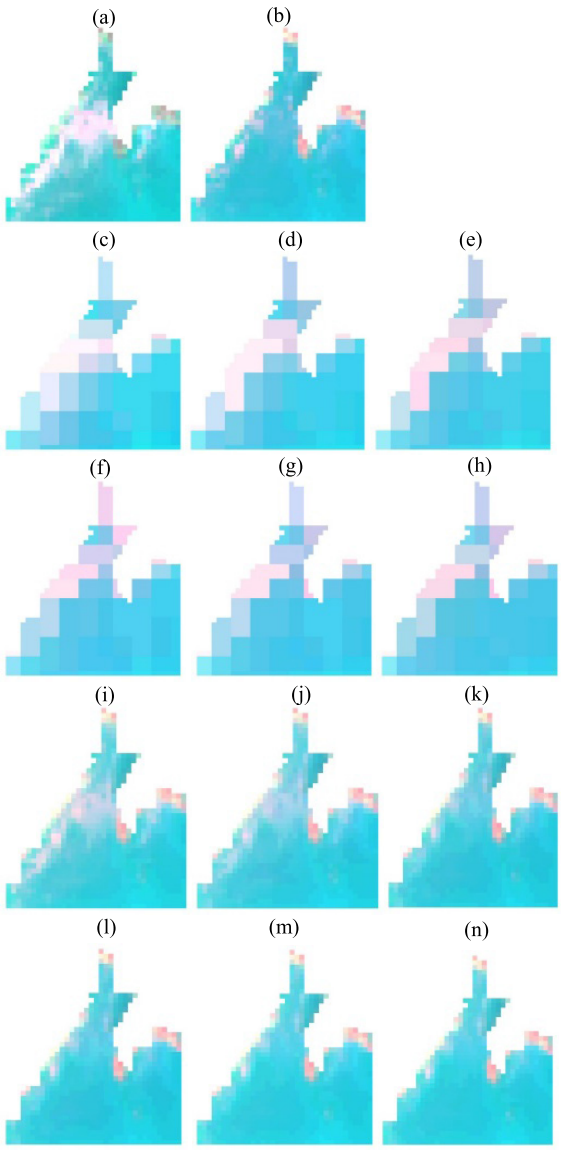


Fig. 17. GOCI images (R: band 7, G: band 4, B: band 3) (a) at 9:30 and (b) 10:30. Himawari images (R: band 4, G: band 3, B: band 2) at (c) 9:30, (d) 9:40, (e) 9:50, (f) 10:00, (g) 10:10, and (h) 10:20. MTIF fusions (R: band 7, G: band 4, B: band 3) at (i) 9:30, (j) 9:40, (k) 9:50, (l) 10:00, (m) 10:10, and (n) 10:20.

the blooming region changed gradually from red to blue, which indicated that Himawari's 10-min temporal resolution could offer detailed temporally changing information for the biophysical processes. However, Himawari's 2-km spatial resolution made it difficult to recognize the bloom region's boundaries and its spatial distribution. In the MTIF fusions [see Fig. 17(i)–(n)], both the changing strength and spatial distribution of the bloom region were clearly observed: The large bloom region at the mouth of Zhushan Bay gradually floated to the north and vanished by 10:00, whereas traces of the bloom stayed in the nearshore water.

C. Assumption of the MTIF Algorithm

For STARFM-like temporal fusion algorithms, the biggest obstacle is to deal with the assumption that "the land cover

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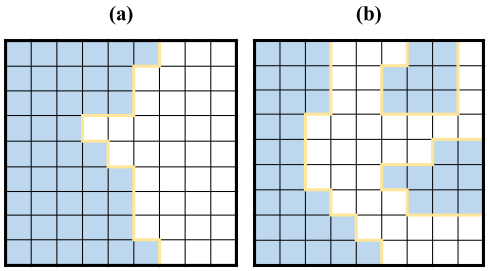


Fig. 18. Simulated high-resolution boundary information within coarse pixel (a) patch #1 and (b) patch #2.

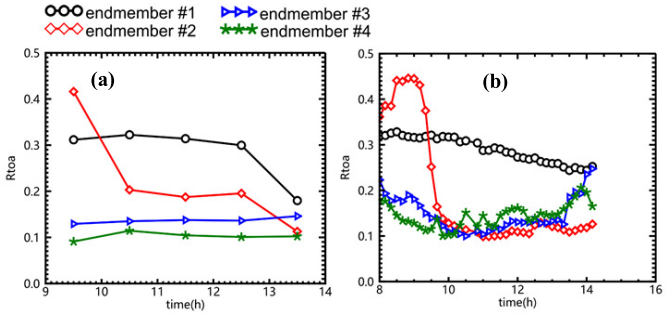


Fig. 19. Temporal endmembers of (a) GOCI and (b) step 1 fused band 7 images of Taihu Lake on July 22, 2016.

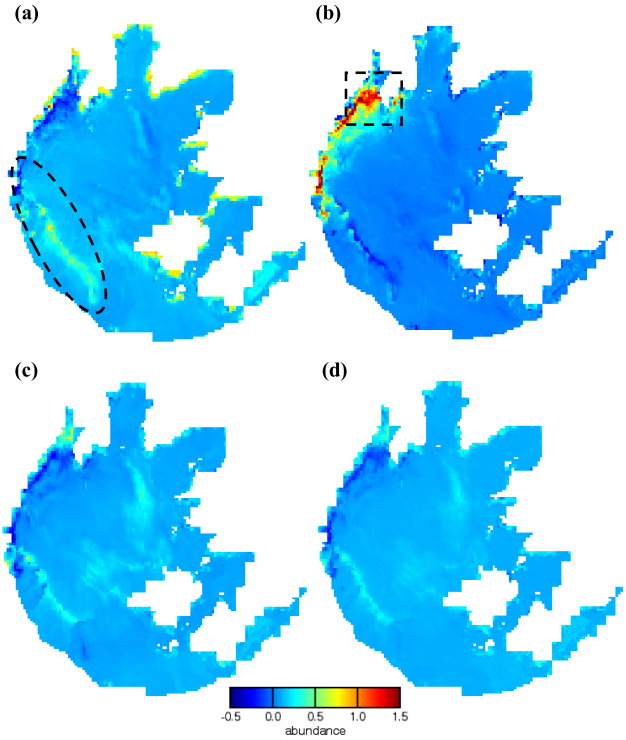


Fig. 20. Temporal abundance images of GOCI band 6 images'. (a) Temporal endmember 1. (b) Temporal endmember 2. (c) Temporal endmember 3. (d) Temporal endmember 4 on July 22, 2016.

type and sensor calibration do not change between the prior and predicted date" [69]. For the region where the land cover changing process was recorded by the coarse images, like the algal bloom in Fig. 17, but the boundary of the bloom region is blurred due to coarse resolution, some spatiotemporal data fusion methods have been developed to improve the fusion results [70]. However, when there are only one or two pairs of images joined in the fusion, the high spatial resolution images can only offer static spatial details for the fusions. Thus, the fusion process was separated into boundary capturing and spectral information estimating [29]. Without any prior knowledge, it is difficult to estimate boundary information from low spatial resolution images if the land cover is changed: Even for two coarse pixels that have the same reflectance value [Fig. 18(a) and (b)], the fine resolution images' land cover boundaries (yellow lines) could be totally different from each other.

Essentially, both the landscape boundaries and the spectrum curves are a combination of the reflectance information: the spatial variance of reflectance information forms landscape boundaries, and the spectral variance of reflectance information forms landscape reflectance spectra. In the MTIF algorithm, one core assumption is that the temporally changing patterns of different image series at the same position are the same. Instead of boundary and spectral information, only the high spatial resolution pixels' changing patterns are considered in the MTIF algorithm. Then, a question about the reliability of the MTIF algorithm needs to be asked: Could the temporal patterns really be analyzed to understand the complex changing water optical properties?

The fast-changing algal bloom conditions in Taihu lake on July 22, 2016 are a good example to answer that question. To this end, the GOCI and step 1 fusion images' temporal endmembers of a pigment-sensitive band (band 7) are shown in Fig. 19. Their corresponding abundance images are shown in Fig. 20. Similar to those in Fig. 15, GOCI and step 1 fusion images' temporal endmembers show good consistency. Four endmembers represent different trends: endmembers 1 and 2 are main contributors to the temporal pattern of the image, and endmember 1 stays stable, at around 0.3. The reflectance value gradually decreases to about 0.2 by 14:00. Different from endmember 1, endmember 2 describes the trend that the reflectance rapidly decreases before 10:00 and then stays at a low contribution level. The GOCI temporal endmember 2 lost the temporal details before 10:00 [see Fig.19(a)]. Endmembers 3 and 4 are secondary contributors to the temporal patterns because their reflectance values are generally lower (around 0.1) and the temporal details are clearer [see endmember 4 in Fig. 19(b)]. This means endmembers 3 and 4 mainly reflect the background temporal pattern and some random situations.

The spatial distribution of the abundance images (see Fig. 20) that correspond to the four endmembers tell more. Abundance image of endmember 1 [see Fig. 20(a)] shows the high abundance pixels distributed in water–land boundaries and the region marked with a black ellipse, which means the reflectance of band 7 in those regions are steady temporally. The GOCI and fusion images in Fig. 14 show that the steady regions on the water–land boundary are influenced by the vegetation pixels in land. The stripped bloom in the southwest region of the lake is the main reason for the high value in the black ellipse in Fig. 20(a). For the abundance image of endmember 2 [see Fig. 20(b)], a small region of large

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abundance value was centralized at the mouth of Zhushan Bay (black rectangle marked region). Abundance in other parts of the image are nearly 0. The severe decrease in the pigment-sensitive band indicates a fierce changing of water properties: This region overlaps with the rapidly changing algal bloom region (see Fig. 14). Combining the abundance map in the region marked with a black rectangle in Fig. 20(b) and the shape of endmember 2, the temporal and spatial detail of the algal bloom's vanishing in Zhushan Bay can be explained well.

V. Conclusion

A spectral–temporal-unmixing-based image-fusion algorithm, MTIF was proposed in this article to fuse high-frequency geostationary satellite images (i.e., Himawari and GOCI in this article). The fusions were quantitatively and qualitatively assessed. The conclusions are as follows.

1) Both the visual interpretation and assessment indexes indicate that the MTIF algorithm has reliable performance. The accuracy of the MTIF algorithm is better than the s1GS, s1PC, s1NND, and s1UBF algorithms. The MTIF algorithm is not sensitive to the outlier points on the land–water edge regions.
2) The C_{chla} maps generated by MTIF, s1, s1PC, s1GS, s1NND, and s1UBF fusions are compared with those by the quasi synchronization GOCI images. The results show that the MTIF and s1UBF C_{chla} maps have better spatial details compared with other algorithms, and MTIF, s1PC, and s1NND C_{chla} maps have better temporal trends compared with other algorithms. What is more, the MTIF algorithm can control the salt-and-pepper noises on the GOCI C_{chla} maps. Overall, MTIF algorithm has the best potential in generating C_{chla} maps.
3) By including the temporal unmixing process, the MTIF algorithm could estimate the reflectance information under cloudy pixels. The experimental image samples indicated that, in pixels where more than 32.6% of usable clean scenes can be found, the MTIF algorithm can successfully recover the undercloud information.
4) By combining the spatial resolution of GOCI images and temporal resolution of Himawari images, the fusions could offer more proper data sources to monitor the rapidly changing water bio-optical properties and biophysical processes.

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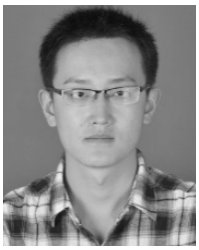
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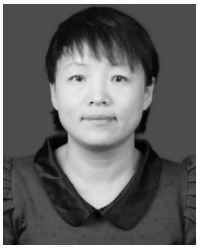
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